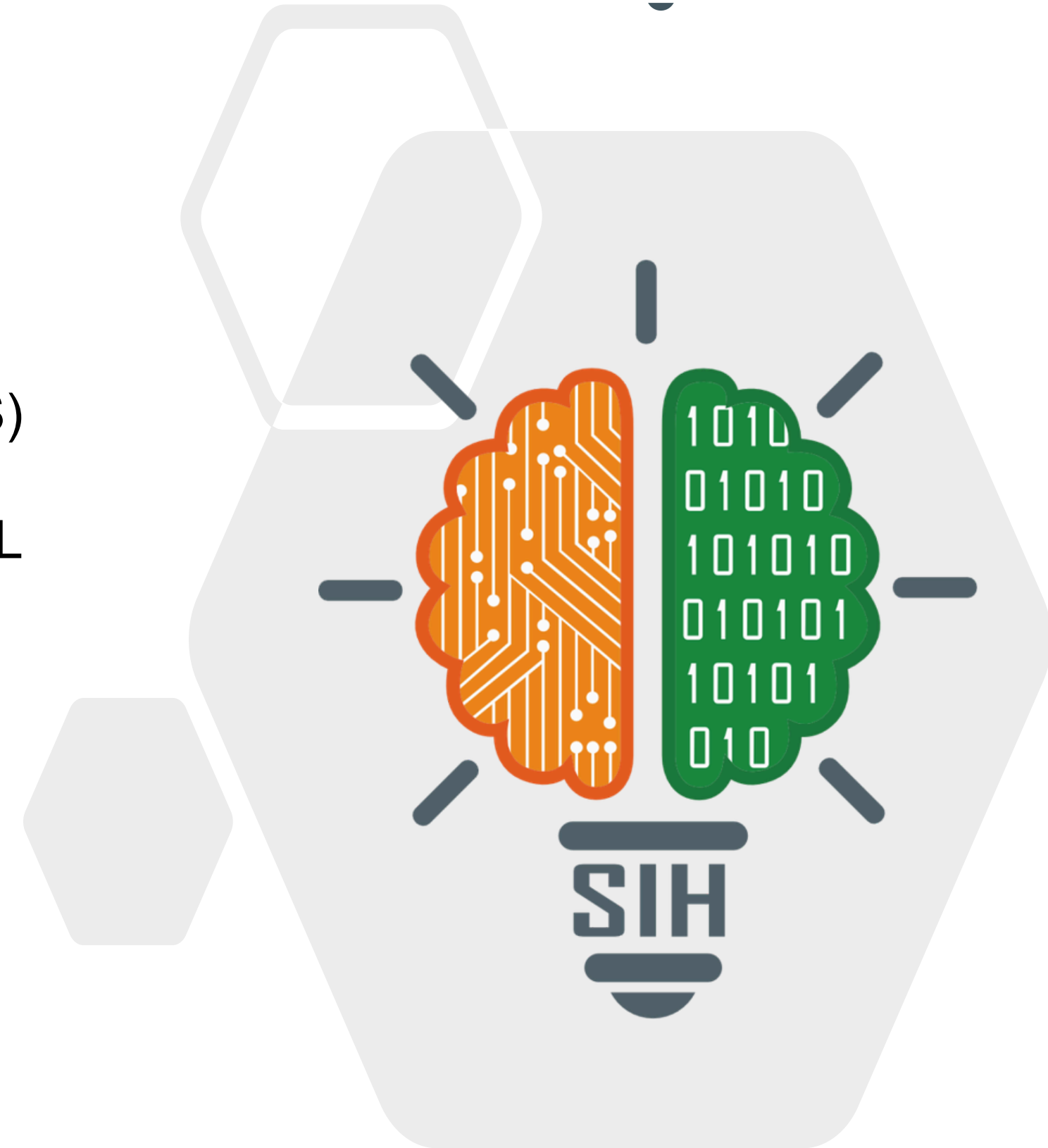


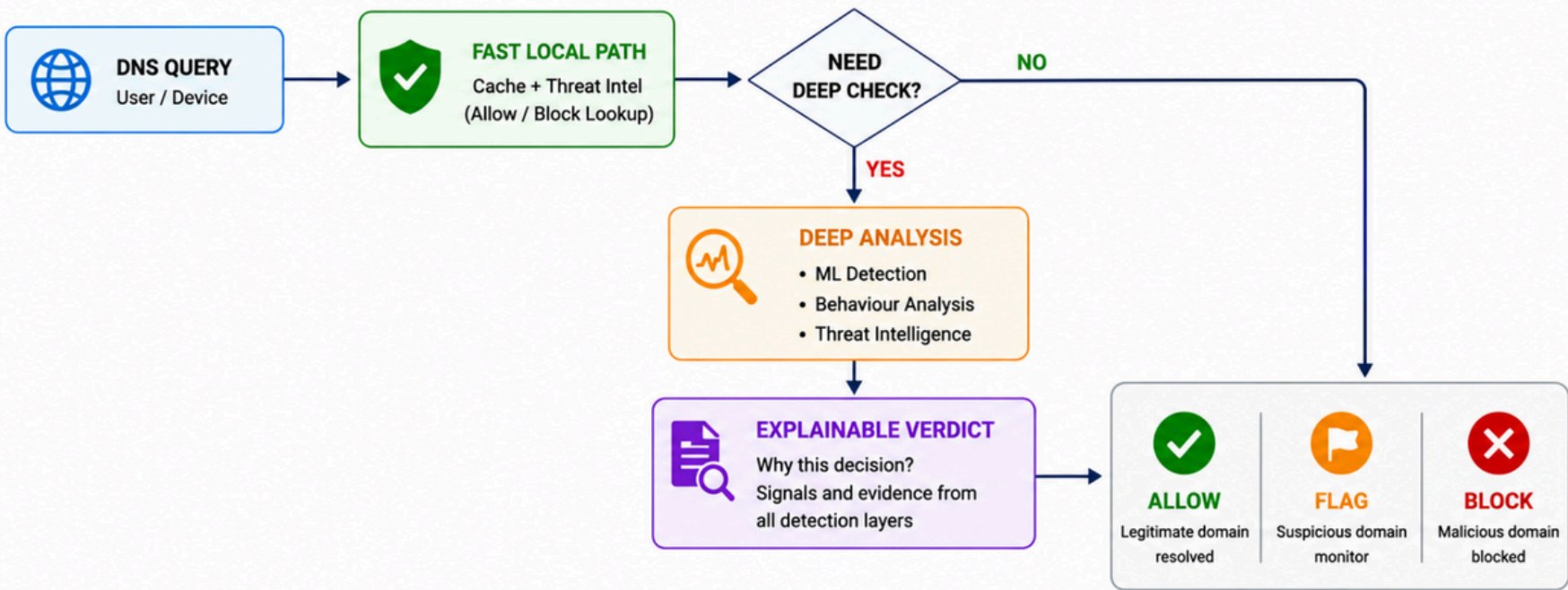
DNS SHIELD

- **Problem Statement ID** – SIH260003
- **Problem Statement Title**- Domain Name Server (DNS)
Filtering Service using Threat Intelligence feeds and AI/ML
Techniques
- **Theme** - Space Technology
- **PS Category**- Software
- **Team ID**-
- **Team Name** - Tragic Bytes



Proposed Solution

- DNS Shield is an intelligent DNS security gateway that filters malicious domains using Threat Intelligence, AI/ML and behavioural analysis while maintaining a fast local resolution path.
- Every DNS query follows a layered decision process:
- Known threats → Threat Intelligence / Cache
- Unknown domains → ML-based detection
- Suspicious behaviour → Device behavioural analysis
- Final result → ALLOW / FLAG / BLOCK



How It Addresses the Problem

PROBLEM		DNS SHIELD RESPONSE
Known malicious domains	→	Threat Intelligence + Cache
New / DGA domains	→	ML-based domain detection
DNS tunnelling	→	Behavioural analysis
ML evasion	→	Adversarial hardening
Threat-feed connectivity	→	Local / offline protection
Unclear security verdicts	→	Explainable ALLOW / FLAG / BLOCK

INNOVATION & UNIQUENESS

OFFLINE-FIRST SECURITY

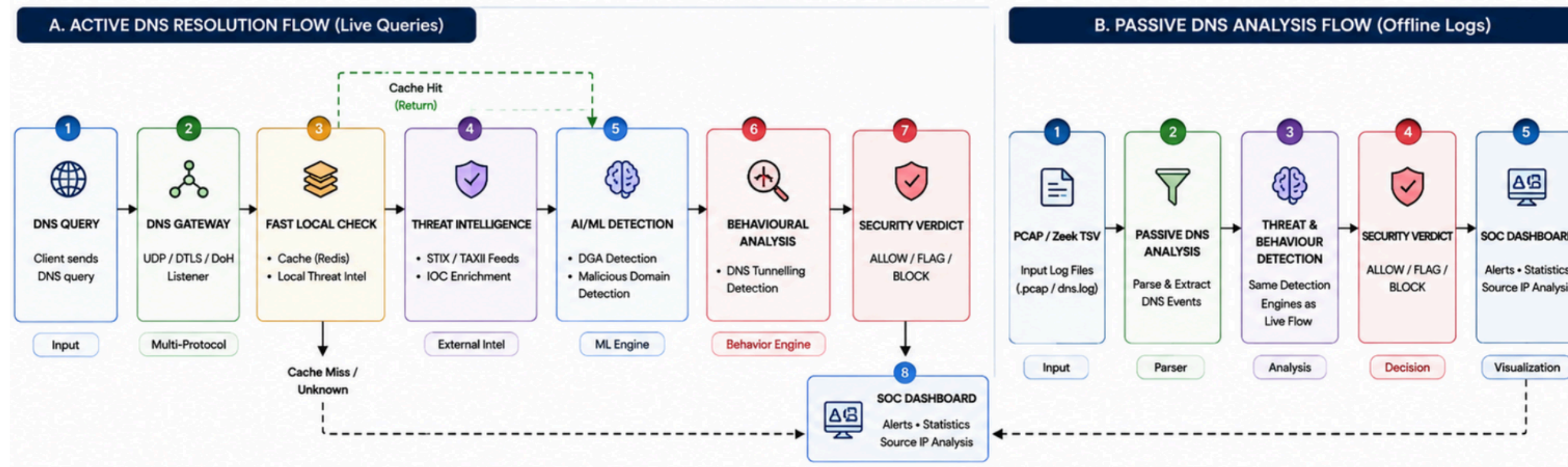
Protection continues locally even when external threat-intelligence connectivity is unavailable, with queued synchronization when connectivity returns.

ADVERSARIALLY HARDENED ML

The ML detector is deliberately tested against evasive domain mutations and retrained on its blind spots.

EXPLAINABLE SECURITY

Every verdict provides an analyst-readable trace showing which detection layers contributed to the decision.



TECHNOLOGIES TO BE USED

BACKEND & DNS

Python 3.11 + FastAPI

DNS gateway,
query processing
and
security APIs

AI / ML

scikit-learn +
Random Forest

Malicious-domain
and
DGA detection

DATA & PERFORMANCE

Redis + ClickHouse

Fast DNS caching
and
security-event analytics

FRONTEND

Next.js 16

SOC dashboard
and
security monitoring

DNS Protocols

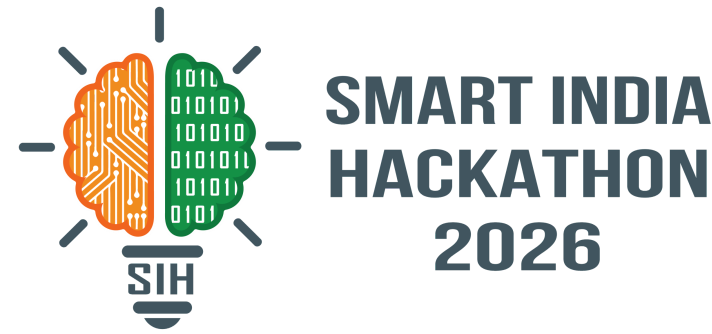
UDP • DTLS • DNS over HTTPS

Input Sources

Live DNS Queries • PCAP • Zeek TSV



FEASIBILITY AND VIABILITY



Built to **scale**. **Hardened** against evasion. **Resilient** when dependencies fail.

01 SCALE & PERFORMANCE

7 Independent Services

Body: Modular FastAPI microservices: Cache, Threat Intel, ML, Behavioral, Geo, Response & Analytics.

Independent Scaling

Compute-heavy services (e.g., ML) can be scaled independently from lightweight services.

Measured Latency

End-to-end query evaluation time measured in local environment.

Latency (local test)

~2.85 seconds per query

Observed under real pipeline load

02 XAI PIPELINE (ARCHITECTURE FLOW)

Stage-wise evaluation with status, risk contribution & reason

1		Cache (Redis)	Check in cache for fast allow	PASS	+0
2		Threat Intel (STIX, URLhaus)	Match IOCs from feeds	HIT	+100
3		Local Rules (9 Rules)	Deterministic local checks	FLAG	+25
4		ML Lexical (Random Forest)	Domain risk score	HIGH	+45
5		Behavioral Engine (Anomaly Detection)	Query pattern analysis	HIGH	+30
6		Geo Intel (IP/ASN/Country)	Geo risk evaluation	MEDIUM	+10
7		Active Response (Block / Sinkhole)	Take action if malicious	BLOCKED	-
8		Analytics Store (ClickHouse)	Log & store for analysis	LOGGED	-

Every verdict is explained with stage status, risk contribution and reason.

Improves analyst trust & faster triage.

Final Verdict

ALLOW

FLAG

BLOCK

03 DEPENDENCY RESILIENCE

Works When Services Fail

If ML, Redis or Threat Intel is down, pipeline continues with available components.

9 Local Rules

Deterministic rules detect suspicious domains even when external feeds or services are unavailable.

Persistent IOC Cache (Disk)

IOC data is stored locally on disk. Survives Redis restarts and network outages.

Graceful Degradation

Service failures do not stop the system. Protection continues with reduced context.

Tested Pipeline

Core pipeline scenarios validated through test plan.

Service Independence

Loose coupling allows services to restart or scale without affecting others.

Offline First Design

Local rules + cached intelligence ensure core protection without external connectivity.

Modular & Maintainable

Clean FastAPI microservices structure for easy updates and future extensibility.



Emerging Threat Protection

Detects DGA, typosquatting and malicious domains in real-time – even against threats not yet present in traditional blocklists.



Protects users, enterprises & critical systems from emerging Internet threats.



Data Loss Prevention

Behavioral analysis detects DNS tunnelling, high-entropy domains, abnormal query volumes and suspicious domain fan-out.



Prevents data leaks, protects IP, and strengthens organizational security.



Analyst Trust via XAI

Every verdict is explained through stage-wise risk contributions, showing exactly why a domain was ALLOWED, FLAGGED or BLOCKED.



Builds trust and speeds up SOC analyst investigations and response.



Sovereign & Offline Capable

9 deterministic local rules and a persistent IOC cache keep threat detection working when external feeds or services are unavailable.



Ideal for government, defense and air-gapped enterprise environments.



~2.85 s

End-to-end decision per DNS query (Demonstrated)



High Accuracy

Adversarially hardened ML reduces false negatives (Evasion tested)



Scalable Architecture

7 independent microservices scale as per workload (Modular & Future Ready)



Resilient by Design

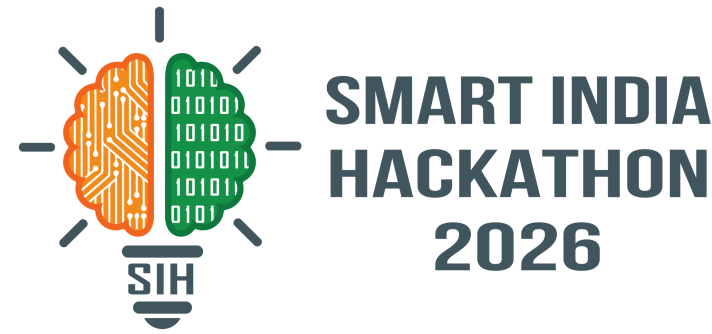
Graceful degradation & local fallback ensure continuity (Always On Protection)



Empowering a Secure, Explainable and Future-Ready Internet for Bharat



RESEARCH AND REFERENCES



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- 1. Mockapetris, P. — Domain Names: Concepts & Facilities, RFC 1034, IETF.
- 2. Mockapetris, P. — Domain Names: Implementation & Specification, RFC 1035, IETF.

THREAT INTELLIGENCE

- 3. OASIS CTI TC — STIX™ Version 2.1.
- 4. abuse.ch — URLhaus Community API Documentation.

DNS SECURITY

- 5. MITRE ATT&CK — DNS: T1071.004.
- 6. Paxson et al. — Practical Comprehensive Bounds on Surreptitious Communication over DNS, USENIX Security.

ML / XAI / ROBUSTNESS

- 7. Woodbridge et al. — Predicting Domain Generation Algorithms with LSTM Networks, 2016.
- 8. Piras et al. — Explaining Machine Learning DGA Detectors from DNS Traffic Data, 2022.
- 9. Vassilev et al. — Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations, NIST, 2025.

PROJECT DOCUMENTATION

[DNS Shield — GitHub Repository](#) · [Architecture](#) · [Project Report](#) · [Enhancement Notes](#)